

Algorithm 1 — §V-B Adaptive Sampling (Exqutor) — B1 baseline 의사코드

본 doc 은 reviewer attack defense B1 (5/10 handoff_v0 §5.4) 대응 — Phase F B1 baseline 의 §V-B Adaptive Sampling 식 1~6 을 의사코드로 명시. Exqutor 본 논문을 그대로 구현했는지 reviewer 가 확인 가능하도록 제공.

Source: Exqutor: Extended Query Optimizer for Vector-augmented Analytical Queries (arXiv:2512.09695v2) §V-B + 식 (1)~(6)

Algorithm 1: Adaptive Sampling for vector cardinality estimation

Inputs:

$q \in \mathbb{R}^d$ — query vector embedding
 $\tau \in \mathbb{R}_+$ — distance threshold (target selectivity)
 $T = \{x_i\}_{i=1..N}$ — table of vector embeddings
Hyperparameters: $m, \eta_0, \alpha, \beta, \gamma, P, N_0, N_{\max}, \epsilon$

Outputs:

$\hat{\mu}_n$ — point estimate of selectivity $p^* = E_x[1\{d(q,x) \leq \tau\}]$
 $\hat{\sigma}^2_n$ — estimated variance
 n — final sample size

Steps:

1. $S \leftarrow$ Bernoulli sample of size N_0 from T (unstratified) $\leftarrow$ Eq. (1)
2. $\hat{\mu}_0 \leftarrow (1/|S|) \sum_{x \in S} 1\{d(q,x) \leq \tau\}$ $\leftarrow$ Eq. (2)
3. $\hat{\sigma}^2_0 \leftarrow \hat{\mu}_0(1 - \hat{\mu}_0)$ $\leftarrow$ Bernoulli variance
4. $v_0 \leftarrow 0, t \leftarrow 0$
5. while convergence not met AND $n < N_{\max}$: $\leftarrow$ Eq. (3) loop
6. $\Delta_t \leftarrow (\hat{\mu}_t - \hat{\mu}_{t-1}) / (|\hat{\mu}_{t-1}| + \epsilon_{\text{safe}})$ (relative grad)
7. $v_t \leftarrow m \cdot v_{t-1} + (1 - m) \cdot \Delta_t$ $\leftarrow$ Eq. (4) momentum
8. $\eta_t \leftarrow \eta_0 \cdot \gamma^{\lfloor t / P \rfloor}$ $\leftarrow$ Eq. (5) lr decay
9. $\hat{\mu}_t \leftarrow \hat{\mu}_{t-1} + \eta_t \cdot v_t$ $\leftarrow$ Eq. (5) update
10. $n_{\text{inc}} \leftarrow N_0 \cdot \max(1, \beta \cdot \hat{\sigma}^2_t / \alpha)$ $\leftarrow$ Eq. (6) batch size
11. Sample n_{inc} more rows via Bernoulli, refresh $\hat{\mu}_t$ and $\hat{\sigma}^2_t$
12. if $t \bmod P == 0$: check $|\hat{\mu}_t - \hat{\mu}_{t-1}| / \hat{\mu}_{t-1} < \epsilon \Rightarrow$ break
13. $t \leftarrow t + 1$
14. return $\hat{\mu}_n, \hat{\sigma}^2_n, n$

Notes: - **Unstratified Bernoulli** (Step 1, 11) — i.i.d. random sampling without distribution-aware partitioning. This is the precise gap our distribution-aware stratification (Phase F B4) augments. - **KNN scope only** — Exqutor §V-B explicitly limits Algorithm 1 to single-table KNN queries (“specifically for KNN queries”). Multi-table joint

distribution (our extended scope) is *not* addressed. – $N_0 = 385$ comes from Cochran (1977) §4.5 sample-size table at 95% confidence $\pm 5\%$ margin for a binomial proportion — i.e., the minimum sample size that yields $\pm 5\%$ half-width 95% CI under unstratified Bernoulli regardless of N (as long as $N \gg 385$).

Hyperparameter table (Exqutor §V-B values, our B1 baseline matches verbatim)

Symbol	Name	Value	Reference	Comment
m	momentum	0.9	Exqutor §V-B Eq.4	matches Adam-style first-moment estimator
η_0	initial learning rate	0.1	Exqutor §V-B Eq.5	base rate before exponential decay
α	curriculum bandwidth	50	Exqutor §V-B Eq.6	controls batch-size growth rate vs σ^2
β	variance penalty	1.5	Exqutor §V-B Eq.6	upweights high-variance regions
γ	lr decay	0.99	Exqutor §V-B Eq.5	per-period decay ($\gamma^{t/P}$)
P	update period	50	Exqutor §V-B Eq.5	queries between lr / convergence checks
N_0	initial sample size	385	Cochran 1977 §4.5	95%/±5% binomial half-width
$N_{\max}$	max sample size	5000	our cap (resource)	empirically suffices for sel ≥ 0.01 in our cells
ϵ	convergence tol	1e-3	Exqutor §V-B	relative change threshold

Implementation cross-reference

Algorithm 1 step	Code reference
Step 1 (Bernoulli S)	<code>_internal/scripts/run_adaptive_sampling.py</code> — <code>bernoulli_init</code>
Step 2 (initial μ_0)	same — <code>mean_indicator()</code>
Step 4 (momentum init)	same — <code>velocity = 0.0</code>
Step 7 (momentum update)	same — <code>velocity = m*velocity + (1-m)*grad</code>
Step 8 (lr decay)	same — <code>lr = lr0 * (gamma ** (t // period))</code>
Step 9 (μ update)	same — <code>mu += lr * velocity</code>
Step 10 (batch size)	same — <code>n_inc = N0 * max(1, beta * var / alpha)</code>
Step 12 (convergence check)	same — <code>!mu_t - mu_prev! / mu_prev < tol</code>

Server-side measurement script: `cache/rq3/run_adaptive_sampling.py` (B1 baseline). All 9 hyperparameters above are exposed as CLI flags (`--momentum` , `--lr0` , `--alpha` , `--beta` , `--gamma` , `--update-period` , `--init-N` , `--max-N` , `--tol`) — measurement reproducibility verifiable by reading parquet `meta.json` .

Why this matters (reviewer attack defense)

Without Algorithm 1 box, a reviewer can challenge:

"Did you implement Exqutor §V-B *exactly* as described? Without an algorithm box matching the paper's equations 1-6, your B1 baseline could differ in subtle ways (e.g., variance estimator, lr schedule, convergence criterion) — invalidating the paired comparison."

This box answers each line of the equations 1-6 with code reference (table above), reproducible hyperparameter values matching the paper verbatim, and explicit mention of *which scope* the algorithm covers (single-table KNN only — our extension is the multi-table generalization).

END

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(`run_adaptive_sampling.py`)